

SIMATS ENGINEERING
SAVEETHA INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES
CHENNAI – 602105
Department of Computer Science and Engineering

ASSESSMENT TOOL 3 – Debugging Test

Course Code:	DSA05	Course Title:	Query Processing for Data Science With Real Time Applications
CO Assessed:	CO2	Assessment:	Tool 3 – Debugging Test
Weightage:		Total Marks:	25 Marks
CO2 Statement: Use Python basics and SQL libraries to connect to databases SQLite, MySQL, PostgreSQL and perform CRUD operations like Create, Read, Update, Delete. (BL6)			
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Section / Batch:	D / 2024	Date of Submission:	06/08/2026
Faculty In-charge:	Dr. B.SHYAMALA GOWRI	Project Mode:	Individual Submission

Code of Conduct

I BHARANI DHARAN S (Name / Reg No) certify that this submission is my original work and that I have adhered to all guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature: bharani S

Error Identification(5)	Code Correction(8)	SQL Query Accuracy(5)	Explanation and Justification(4)	Program Execution and Output(3)	Total (25 Marks)

```

Q4. Debug the PostgreSQL Program
import psycopg2
conn=psycopg2.connect(dbname="college",user="postgres",password="admin"
)
cursor=conn.cursor()
cursor.execute("INSERT INTO Student VALUES(%s,%s,%s)",(1,"Ram"))
conn.commit()
cursor.execute("SELECT * FROM Student")
print(cursor.fetchone())
conn.close()

```

Task

- Identify all errors.
- Rewrite the corrected program.
- Explain the effect of parameter mismatch.

PROGRAM

```

import psycopg2

# Connect to PostgreSQL database
conn = psycopg2.connect(
    dbname="college",
    user="postgres",
    password="admin"
)

cursor = conn.cursor()

# Insert 3 values for 3 placeholders
cursor.execute(
    "INSERT INTO Student VALUES (%s, %s, %s)",
    (1, "Ram", 20)
)

conn.commit()

# Display inserted record
cursor.execute("SELECT * FROM Student")
print(cursor.fetchone())

cursor.close()
conn.close()

```

Effect of Parameter Mismatch

The SQL statement contains **3 parameters**:

```
INSERT INTO Student VALUES (%s, %s, %s)
```

But only **2 parameters** are provided:

```
(1, "Ram")
```

Therefore, PostgreSQL/Pyocpg2 raises an error because **the number of supplied parameters does not match the number of placeholders**.

Example error:

IndexError: tuple index out of range

Conclusion: The number of %s placeholders must exactly match the number of values supplied.

Q8. Debug the Database Fetch Program `cursor.execute("SELECT \* FROM Student")`

```
rows=cursor.fetchall()
```

```
print(rows[20])
```

```
cursor.close()
```

```
cursor.execute("SELECT \* FROM Student")
```

****Task****

- Identify runtime errors.
- Explain why the program fails.
- Rewrite the corrected program.

PROGRAM :

```
cursor.execute("SELECT * FROM Student")
```

```
rows = cursor.fetchall()
```

```
# Check whether rows are available
```

```
if len(rows) > 20:
```

```
    print(rows[20])
```

```
else:
```

```
    print("Less than 21 student records found.")
```

```
# Execute all required queries before closing the cursor
```

```
cursor.execute("SELECT * FROM Student")
```

```
rows = cursor.fetchall()
```

```
print(rows)
```

`cursor.close()`

Result

- If 21 or more records exist, `rows[20]` displays the **21st student record**.
- If fewer than 21 records exist, it displays:

Less than 21 student records found.

- The cursor is closed **only after completing all database operations**.